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https://github.com/LeonVir/NLU

Authorship Verification via Stylometric Ensemble & RoBERTa-Large Fine-tuning

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Abstract

Authorship Verification (AV) is a fundamental binary NLU task that determines whether two distinct texts share the same author. It plays a critical role in forensic linguistics, plagiarism detection, and combating digital misinformation. In this study, we tackle the inherent difficulties of cross-topic authorship attribution by presenting Group 33's two complementary approaches:

Model A: Stylometric Feature Ensemble
High-efficiency pipeline with no neural components.
Macro F1 = $\boxed{0.7702}$

Model C: RoBERTa-Large + Asymmetric Loss (ASL)
Robust fine-tuning to handle mixed-domain overfitting.
Macro F1 = $\boxed{0.8348}$ (+0.0646 over Model A)

Introduction

Problem Definition: Given a pair of texts (s_1, s_2), the objective is to predict y=1 (same author) or y=0 (different author). This requires models to effectively isolate an author's intrinsic *stylistic signature* from the semantic topic.

Key Challenges in Real-World AV

- **Stylistic Sparsity:** Short, informal texts provide highly limited lexical and syntactic cues.
- **Domain & Genre Shifting:** An author's writing style naturally fluctuates across contexts (e.g., email vs. chat).
- **Intra-Author Variation:** Individuals exhibit stylistic inconsistencies over time, making static profiling difficult.
- **Closed-Track Restriction:** No external stylistic datasets are permitted, strictly limiting the learning boundary.

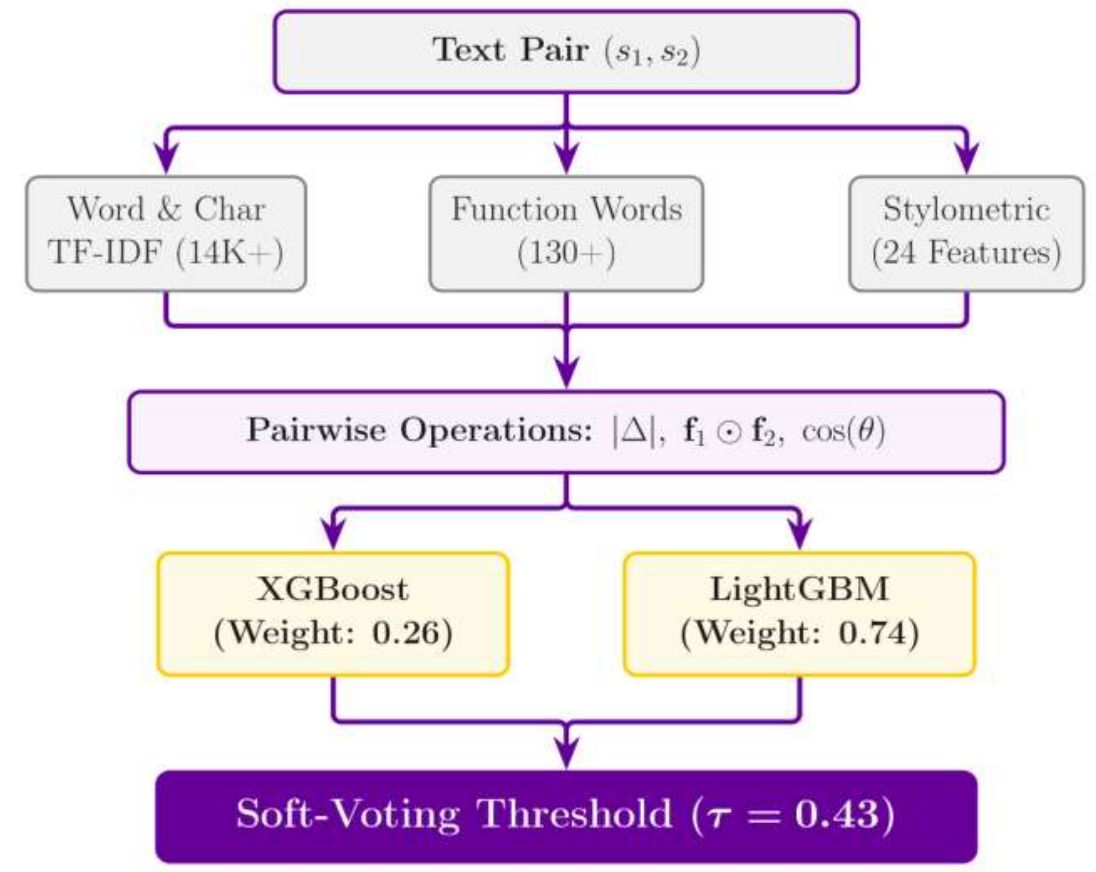
Dataset Overview

The project utilizes a **closed-track** dataset provided for the COMP34812 AV task. It comprises a highly diverse set of real-world texts, primarily drawn from corporate communications, reflecting various lengths and linguistic styles.

Split	Pairs	Note
Train	27,000+	Unlabelled; labels provided separately
Validation	**5,993**	**Dev F1 eval.** (Label 0/1 ≈ 50%)
Test	Hidden	Final submission track

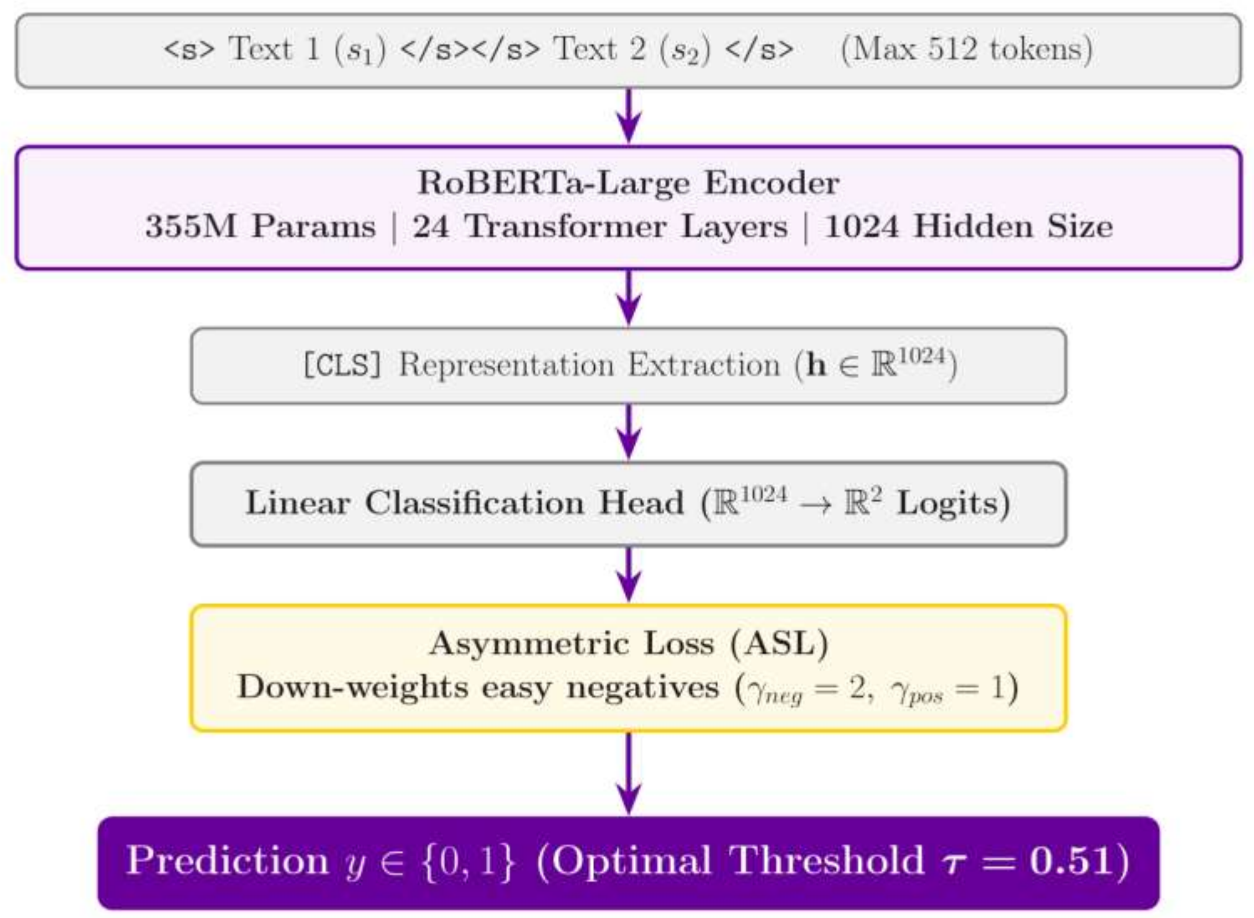
Model A: Stylometric Ensemble

Model A takes a highly engineered, feature-driven Machine Learning approach. By extracting over 14,000 multidimensional stylometric signals, it leverages a robust soft-voting ensemble of gradient boosting models (XGBoost & LightGBM). This pipeline ensures extreme computational efficiency while effectively capturing intricate lexical and syntactic nuances between text pairs.



Model C: RoBERTa-Large & ASL

Model C leverages the immense representational capacity of a 355M-parameter `RoBERTa-Large` encoder to extract deep, context-aware stylistic features. Our core architectural innovation is the integration of **Asymmetric Loss (ASL)**. By dynamically decoupling the penalty for positive and negative classes, ASL aggressively down-weights easily distinguishable negatives. This effectively neutralizes severe domain overfitting and forces the model to focus on the most challenging, ambiguous text pairs.

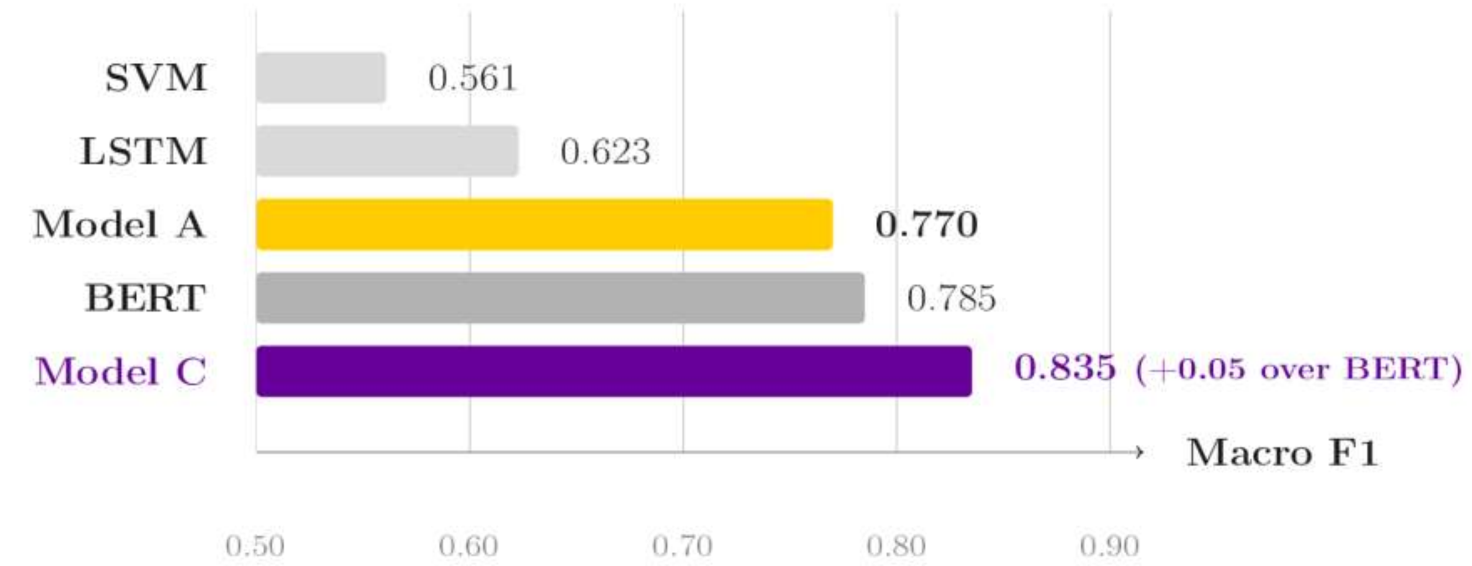


Why ASL over Cross-Entropy (CE)?

CE caused rapid overfitting on easy negatives, leading to an unstable threshold (0.38) and high validation loss (1.12). ASL dynamically down-weights easy examples, yielding a highly stable learning curve (Val Loss 0.27) and a balanced decision threshold ($\tau = 0.51$).

Experimental Results

Model C overwhelmingly surpasses all official baselines, demonstrating the superiority of the ASL-fine-tuned RoBERTa-Large architecture. Model A also shows highly competitive performance given its computational efficiency.



Discussion & Conclusion

Limitations: RoBERTa's 512-token limit truncates trailing stylistic cues in long documents. Model A bypasses this natively, offering a white-box alternative with global document coverage.

Error Analysis:

- **Model A — Input Length:** Handcrafted stylometric features require sufficient text; extremely short pairs (e.g. "I agree") degrade prediction accuracy.
- **Model A — OOV & Feature Over-reliance:** TF-IDF discards unseen vocabulary at test time. Feature importance analysis revealed over-sensitivity to `word_prod` and `char_diff`, producing spurious False Positives/Negatives.
- **Model C — Truncation:** Pairs exceeding 512 tokens lose trailing stylistic cues (sign-offs, closing style), causing misclassification on long emails.

Conclusion: Bridging traditional stylometry (Model A) and deep semantic modelling (Model C) addresses both length constraints and complex stylistic nuances. Integrating **ASL** effectively suppressed domain-shift overfitting, propelling Model C to outperform all baselines.

Ablation Insights & Ethical Considerations

Ablation — Why not DeBERTa-v3? DeBERTa-v3-base yielded sub-optimal performance (Max F1 ≈ 0.77), falling short of even BERT, due to:

- **Objective Clash:** DeBERTa's RTD pre-training conflicted with nuanced stylistic signals under the ASL framework.
- **Domain Mismatch:** DeBERTa's corpus aligns less effectively with Enron email structures compared to RoBERTa's byte-level BPE.

Ethical Considerations:

- **Privacy:** AV systems can be misused to de-anonymise authors or whistleblowers in adversarial deployments.
- **Domain Bias:** Training data drawn from corporate communications may underrepresent non-native speakers or informal writing styles, potentially introducing attribution bias.

Reproducibility: All models and training logs are available at `github.com/LeonVir/NLU`